Space of analytic functions over unit disk — Notes

November 8, 2026



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1 Lecture 1 — 18.10.2026

The main object of this course is a Hardy Space, for $0 < p < \infty$, the Hardy space $\mathbb{H}^p$ consists of all holomorphic functions $f : \mathbb{D} \to \mathbb{C}$ such that,

$$\sup_{0<r<1} \int_{-\pi}^{\pi} |f(re^{it})|^p \, dt < \infty$$

and,

$$\mathbb{H}^\infty = \{\text{bounded analytic functions on } \mathbb{D}\}$$

for $p < q$ it is clear that $\mathbb{H}^p \supseteq \mathbb{H}^q \supseteq \ldots \supseteq BMO \supseteq \mathbb{H}^\infty$. We study boundary behavior, in particular Fatou's theory; for a.e. $\theta \in [0, 2\pi)$, a function $f \in \mathbb{H}^\infty$ has a non-tangential limit at $e^{i\theta}$,

$$z_n to e^{i\theta} \implies \lim_{n\to\infty} f(z_n) \text{ exists}$$

in particular $\lim_{r\to 1^-} f(re^{i\theta})$ exists.

Another question we study is Zero sets, a set of points $\{a_n\} \subseteq \mathbb{D}$ is the Zero set of an $\mathbb{H}^\infty$ function if and only if $\sum(1 - |a_n|) < \infty$, in other words it converges to the boundary quickly (Blaschke condition).

We also study Invariant subspaces $zX \subseteq X$, if $z(z) \in X$ then $z \cdot f(z) \in X$ (Beurling, 1949).

Interpolating sequences. A sequence $\{a_n\} \subseteq \mathbb{D}$ is interpolating if for any $b_n \in l^\infty$ there exists $f \in \mathbb{H}^\infty$ such that $f(a_n) = b_n$. The answer was given by Carleson: a sequence is interpolating if and only if it is separated and has Carleson condition.

Finally, Corona problem: given $f_1, f_2 \in \mathbb{H}^\infty$ such that $|f_1(z)| + |f_2(z)| > \delta$ for all $z \in \mathbb{D}$, then there are $g_1, g_2 \in \mathbb{H}^\infty$ such that $f_1 g_1 + f_2 g_2 = 1$. The Space BMO is going to be involved with the solution to this problem. There are also Bergman spaces, $A^p = \{f : \mathbb{D} \xrightarrow{\text{hol}} \mathbb{C} \mid \int_{\mathbb{D}} |f(z)|^p \, dA < \infty\}$ and Dirichlet space,

$$\mathcal{D} = \{f : \mathbb{D} \xrightarrow{\text{hol}} \mathbb{C} \mid \int_{\mathbb{D}} |f'(z)|^2 \, dA < \infty\}$$

The textbook is "H_p spaces" by P. Koosis. The preliminaries are Complex analysis 1, Measure theory, Hilbert spaces and little bit of Functional analysis.

Reminder about $\mathcal{L}^p$ spaces, $(\mathcal{L}^p)^* \simeq \mathcal{L}^q$ for $1 \le p < \infty$ where $\frac{1}{p} + \frac{1}{q} = 1$. Then $\|g\|_{L^q} = \sup_{\|f\|_{L^p}} \int f \cdot g \, dt$. Another fact is that $C_c(\mathbb{R}) \subseteq \mathcal{L}^p(\mathbb{R})$ is dense. $(T_t f)(x) = f(x + t)$ is continuous in $\mathcal{L}^p$ in the sense that $t \to T_t f$ is continuous,

$$\|T_t f - f\|_{L^p} \xrightarrow{t\to 0} 0$$

the proof is left as exercise. $C(\partial\mathbb{D})^* \simeq \mathcal{M}(\partial\mathbb{D})$ and $C_0(\mathbb{R})^* \simeq \mathcal{M}(\mathbb{R})$. We can define the measure μ that takes $f \to \int_{\partial\mathbb{D}} f \, d\mu$.

1.1 Harmonic functions

Definition 1.1 (Harmonic function)**.** a function $u : \Omega \to \mathbb{R}$ is harmonic if one of the following conditions holds:

1. u is C^2 and $\Delta u = \frac{\partial^2}{\partial x^2} u + \frac{\partial^2}{\partial y^2} = 0$
2. $u \in C$ satisfies the mean-value property, namely for any disk $B(z_0, r) \subseteq \Omega$ it holds that,
$$u(z_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(z_0 + re^{i\theta}) \, d\theta$$
3. u is locally the real part of an analytic function.

Example 1.1. $\log|z|$ is harmonic in $\mathbb{C} \setminus \{0\}$.

We would have wanted to take $f = \log z$ but we would've encountered a problem with continuity. Note that condition 3 holds for the imaginary part as well. For the sake of convenience we will treat condition 3 as the main definition in this

course. Suppose that $u : B(0,R) \to \mathbb{R}$ is harmonic, then $u = \text{Re}\, f$ for some analytic function f, but any such function has an analytic series,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

such that,

$$\liminf_{n\to\infty} \frac{1}{\sqrt[n]{|a_n|}} \geq R$$

Taking real parts, we get,

$$u(rr^{i\theta}) = \sum_{n=-\infty}^{\infty} A_n r^{|n|} e^{in\theta}$$

In order for this to be real we need $A_{-n} = \overline{A_n}$ and the condition,

$$\forall n \in \mathbb{Z},\ \liminf_{n\to\infty} \frac{1}{\sqrt[|n|]{|A_n|}} \geq R$$

1.2 Poisson integral formula

Poisson formula gives us some generalization of the mean-value property in the sense that each point is valued by an harmonic function as the boundary of a consisting ball, up to weight function (kernel). Let $u : B(0,R) \to \mathbb{R}$ with $R > 1$ and $r < 1$ then,

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(e^{it}) \left[\sum_{-\infty}^{\infty} r^{|n|} e^{in(\theta - t)} \right] dt$$

We know that,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{int}\, dt = \begin{cases} 0 & n \neq 0 \\ 1 = 0 & \end{cases}$$

and that if u is defined as a sequence, then,

$$u(re^{it}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \sum_{-\infty}^{\infty} A_n e^{int} r^{|n|} e^{in(\theta - t)}\, dt$$

thus we get the exact formula.

Definition 1.2 (Poisson kernel)**.** For any θ we define,

$$P_r(\theta) = \sum_{-\infty}^{\infty} r^{|n|} e^{in\theta} = \frac{1 - r^2}{1 + r^2 - 2r\cos\theta} = \text{Re}\left(\frac{1 + re^{i\theta}}{1 - re^{i\theta}}\right)$$

Then,

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(e^{it}) P_r(\theta - t)\, dt$$

Proposition 1.3. *The following properties hold for Poisson kernel:*

1. $P_r(\theta) > 0$
2. P_r *is* 2π*-periodic*
3. $\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta)\, d\theta = 1$
4. *For any* $\delta > 0$, $P_r(\theta) \to 0$ *uniformly for* $\pi > |\theta| > \delta$, *as* $r \to 1$
5. $P_r(\theta)$ *is even*

Remark. We can also denote using Poisson kernel with,

$$u(er^{i\theta}) = \frac{1}{2\pi} u(e^{it}) P_r(\theta - t)\, dt$$

Suppose that $f \in \mathcal{L}^p(\partial\mathbb{D})$ or equivalently $\in \mathcal{L}^p([-\pi,\pi])$, and let $u = P[f]$ Poisson extension of f, namely,

$$u(re^{it}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)P_r(\theta - t)\,dt$$

then u is harmonic in $\mathbb{D}$. We can also consider $u + i\tilde{u}$ for $\tilde{u}$ the harmonic conjugate. So we can plug $\mathcal{L}^p$ functions into the PIF (Poisson Integration Formula) to obtain harmonic functions in the disk. What kind of functions do we get by doing so, and how are u and f related? Is $f \to u \to \tilde{u} \to \tilde{f}$ bounded in $\mathcal{L}^p$ (Singular integral operators)? That is a delicate question which we will answer during the course.

Theorem 1.4 (Extension to harmonic functions)**.** *Suppose that $u(re^{i\theta})$ is a harmonic function on the unit disk. If $1 < p < \infty$ and,*

$$\sup_{0<r<1}\int_{|z|=r} |u(re^{i\theta})|\,d\theta < \infty$$

then there is a function $f : [-\pi,\pi] \to \mathbb{R}$ in L^p such that $u(re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - t)f(t)\,dt$.

Proof. Compactness: We know that $(\mathcal{L}^q)^* \simeq \mathcal{L}^p$. Choose a countable dense set $G_1, G_2, \ldots \in \mathcal{L}^q$. Define,

$$\mathcal{L}G_j = \lim_{j\to\infty}\frac{1}{2\pi}\int_{-\pi}^{\pi} G_k(\theta)u((1-\frac{1}{n_j})e^{i\theta})\,d\theta$$

$G_k \in \mathcal{L}^q$ and $u((1-n_j)e^{i\theta})$ is bounded because L^p norm is bounded, then the whole integral is bounded. Remember that $|\int fg| \le \|f\|_p \cdot \|g\|_q$ for any $f \in L^p, g \in L^q$. For a general $G \in \mathcal{L}^q$, we approximate G by $G_{k_l} \in \mathcal{L}^q$ and define $\mathcal{L}G = \lim_{l\to\infty}\mathcal{L}G_{k_l}$. It follows that there exists a function $f \in \mathcal{L}^p$ such that $\mathcal{L}G = \int_{-\pi}^{\pi} fG$. Fix $-\pi \le \theta \le \pi$ and choose $G(t) = P_r(\theta - t)$, let $v = P[f]$,

$$\mathcal{L}P_r(\theta - t) = \int_{-\pi}^{\pi} f(t)P_r(\theta - t)\,dt = 2\pi v(re^{i\theta}) = \lim_{j\to\infty}\int_{-\pi}^{\pi} P_r(\theta - t)u((1-\frac{1}{n_j})e^{i\theta})$$

The dilate $u((1-\frac{1}{n_j})e^{i\theta})$ is harmonic on $B(0, \frac{1}{1-\frac{1}{n_j}}) \subseteq B(0,1)$ therefore we can use Poisson formula,

$$\lim_{j\to\infty}\int_{-\pi}^{\pi} P_r(\theta - t)u((1-\frac{1}{n_j})e^{i\theta}) = \lim_{j\to\infty} 2\pi u((1-\frac{1}{n_j})re^{i\theta}) = 2\pi\cdot u(re^{i\theta})$$

□

Theorem 1.5 (Existence of measure to harmonic functions)**.** *Suppose that $u(er^{i\theta})$ is a harmonic function on $\mathbb{D}$ such that,*

$$\sup_{0<r<1}\int_{-\pi}^{\pi} |u(re^{i\theta})|\,d\theta < \infty$$

then there exists a measure μ such that,

$$u(re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - t)\,d\mu$$

Example 1.2. Remember that $P_r(\theta) = \frac{1-r^2}{1+r^2-2\cos\theta}$, in the case "$r = 1$" acts like δ_1.

2 Lecture 2 — 25.10.2026

2.1 Measures and total variation

A **signed measure** is the difference of two positive finite measures, $\mu = \mu_+ - \mu_-$. The **total variation measure** $|\mu| = \mu_+ + \mu_-$, this leads to the identity,

$$\left| \int f \, d\mu \right| \leq \int |f| \, d|\mu|$$

We define $\|\mu\|_M = |\mu|(X)$. A complex measure is, in similar manner, a sum of two measures, $\mu = \lambda + i\nu$ where λ, ν are signed measures. In this case we define the total variation as $|\mu| = |\lambda| + |\nu| = \lambda_- + \lambda_- + \nu_+ + \nu_-$.

Theorem 2.1 (Banach-Alaoglu)**.** *Let X be a Banach space, The unit ball in X^* is weak-$*$ compact.*

Example 2.1. $(\mathcal{L}^q)^* \simeq \mathcal{L}^p$ for $1 <\leq \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$.

Suppose $f_1, f_2, \ldots$ with $\sup\|f_j\|_{L^p} < \infty$. $\{f_j\}$ may not have a continuous subsequence in $\mathcal{L}^p$, BA (Banach-Alaoglu) implies that there are j_k such that $f_{j_k} \to f$ in weak-$*$,

$$\forall g \in \mathcal{L}^q, \quad \int f_{j_k} g \, dx \to \int f g \, dx$$

2.2 Harmonic functions in R-disk

Denote $\mathbb{D}_R = \{z \in \mathbb{C} \mid |z| < R\}$. Function $u : \mathbb{D}_R \to \mathbb{R}$ is harmonic if,

$$u(re^{i\theta}) = \sum_{n=-\infty}^{\infty} A_n r^{|n|} e^{in\theta}$$

where $A_{-n} = \bar{A}_n$ and radius $\geq R$.

We can consider complex-valued harmonic functions, in this case we can omit the first condition and say that u is analytic iff $A_{-n} = 0$ for $n > 0$,

$$u(re^{i\theta}) = \sum_{n\geq 0} A_n r^n e^{in\theta}$$

u is **anti-analytic** iff $A_n = 0$ for $n > 0$. Any harmonic function h can be written as $h = i + iv$ where u, v are real-valued harmonic functions, but also as $h = f + \bar{g}$ where f, g are analytics.

We say that v is a **harmonic-conjugate** of u if $u + iv$ is analytic.

Lemma 2.2. *v is unique up to an additive constant.*

If v_1, v_2 are such conjugates then $(u + iv_1) - (u + iv_2) = i(v_1 - v_2)$ which is constant.

Existence:

$$\tilde{u}(re^{i\theta}) = \sum_{n=0\infty}^{\infty} (-i \operatorname{sgn} n) A_n r^{|n|} e^{in\theta}$$

From this definition it follows that $\tilde{A}_n = -iA_n$ for $n > 0$ and so $\tilde{A}_{-n} = iA_{-n}$, meaning that indeed $\tilde{A}_{-n} = \bar{\tilde{A}}_n$.

When discussing Poisson integral formula we assumed that $R > 1$,

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(e^{it}) \left[\sum_{n=-\infty}^{\infty} r^{|n|} e^{in(\theta - t)} \right] dt$$

When $R > 1$, $|A_n| \leq C\Theta^{-n}$ for some $\Theta > 1$.

Example 2.2. The series $z + z^2 + \cdots$ has a radius $1 = \limsup \sqrt[|n|]{|A_n|}$, to enlarge it we have to set the coefficients to be much smaller. $\frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8}$ has a radius of 2.

We will define Poisson kernel as,

$$P_r(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta} = \frac{1 - r^2}{1 + r^2 - 2r\cos\theta} = \operatorname{Re} \frac{1 + re^{i\theta}}{1 - re^{i\theta}}$$

Poisson kernel $P(re^{i\theta})$ is a harmonic function in $\mathbb{D}$. This is clear as it can be written as a special case of the series defining a harmonic function, or as the real part of an analytic function. In general we can achieve the harmonic conjugate with the formula,

$$\tilde{u}(re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} u(e^{it})\left[\sum_{n=-\infty}^{\infty} -i\,\mathrm{sgn}(n) r^{|n|} e^{in(\theta-t)}\right] dt$$

So we define the conjugate Poisson kernel as,

$$Q_r(\theta) = \sum_{n=-\infty}^{\infty} -i\,\mathrm{sgn}(n) r^{|n|} e^{in\theta} = \frac{2r\sin\theta}{1+r^2-2r\cos\theta} = \mathrm{Im}\,\frac{1+re^{i\theta}}{1-re^{i\theta}}$$

2.3 Harmonic functions with bounded Lp means

Theorem 2.3 (Representation with Lp fuctions)**.** *Suppose u is harmonic function on the unit disk. If,*

$$\sup_{0<r<1}\int_{-\pi}^{\pi} |u(re^{i\theta})|^p\, d\theta < \infty \tag{1}$$

then,

$$u(re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(t)P_r(\theta-t)\, dt$$

where $F \in \mathcal{L}^p$.

Proof. Set $F_n(\theta) = u((1-\frac{1}{n})e^{i\theta})$ for $\theta \in [-\pi,\pi]$. (1) sways that $\|F_n\|_{L^p} \le C$ for all n. Banach-Alaoglu means that for any $G \in L^q$,

$$\int_{-\pi}^{\pi} F_{n_k}(\theta)G(\theta)\, d\theta \to \int_{-\pi}^{\pi} F(\theta)G(\theta)\, d\theta$$

Fix $0 \le r < 1$ and $t \in [-\pi,\pi]$ and take $G(\theta) = P_r(t-\theta)$, this is a smooth function.

$$2\pi u(re^{it}) = 2\pi \lim_{k\to\infty} u((1-\frac{1}{n_k})re^{it}) = \lim_{k\to\infty}\int_{-\pi}^{\pi} u((1-\frac{1}{n_k})e^{i\theta})P_r(t-\theta)\, d\theta = \int_{-\pi}^{\pi} F(\theta)P_r(t-\theta)\, d\theta$$

□

Theorem 2.4 (equivalency in the case of 1)**.** *Suppose u is harmonic function on the unit disk and $p = 1$. If,*

$$\sup_{0<r<1}\int_{-\pi}^{\pi} |u(re^{i\theta})|\, d\theta < \infty$$

then,

$$u(re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta-t)\, d\mu(t)$$

where μ is a signed measure.

Corollary 2.5. Suppose $u : \mathbb{D} \to \mathbb{R}$ is a positive harmonic function. Then $u = P[\mu]$ for μ a positive measure, namely $du = Pd\mu$.

Example 2.3. $P(re^{i\theta}) = \frac{1-r^2}{1+r^2-2r\cos\theta}$ is a positive harmonic function,

$$P(re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta-t)d(2\pi\delta_0(t))$$

It follows that $P = P[2\pi\delta_0]$.

Example 2.4 (Plateau's problem)**.** Suppose that $\gamma \subseteq \mathbb{R}^3$ is a closed smooth path, find a surface $S \subseteq \mathbb{R}^3$ with $\partial S = \gamma$ of minimal area.

Let $m = \inf\{\mathrm{vol}_2\, S \mid \partial S = \gamma\}$, for any $n \in \mathbb{N}$ there is a surface S_n with $\partial S_n = \gamma$ and $\mathrm{vol}_2\, S_n < m + \frac{1}{n}$.

Theorem 2.6. *Suppose $F(\theta) \in \mathcal{L}^p$, where $1 < p \le \infty$, and $u = P[F]$. Then,*

$$\sup_{0<r<1}\int_{-\pi}^{\pi} |u(re^{i\theta})|^p\, d\theta < \infty$$

Proof. Set $F_r(\theta) = u(re^{i\theta})$ for $\theta \in [-\pi, \pi]$. Then there is a function $G \in \mathcal{L}^q$ with $\|G\|_{L^q} = 1$ such that,

$$\begin{aligned}
\|F_r\|_{L^p} &= \int_{-\pi}^{\pi} F_r(\theta)G(\theta)\, d\theta \\
&= \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} F(\theta - t)P_r(t)\, dt\, G(\theta)\, d\theta \right| \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \|F\|_{L^p} \|G\|_{L^q} P_r(t)\, dt \\
&= \|F\|_{L^p} \|G\|_{L^q} \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(t)\, dt \\
&= \|F\|_{L^p} \|G\|_{L^q} \\
&= \|F\|_{L^p}
\end{aligned}$$

In the case $p = 1$, $G \in \mathcal{L}^\infty$ and $G = \frac{\bar{F}_r}{|F_r|}$, the inequalities are a bit different,

$$\begin{aligned}
\|F_r\|_{L^1} &= \int_{-\pi}^{\pi} F_r(\theta)G(\theta)\, d\theta \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} P_r(\theta - t) d\mu(t) G(\theta)\, d\theta \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} P_r(\theta - t) |d\mu(t)|\, d\theta \\
&= \int_{-\pi}^{\pi} |d\mu(\theta)| \\
&= \|\mu\|_M
\end{aligned}$$

and the claim holds. □

2.4 Elementary study of boundary behavior

Theorem 2.7 (Extension of unit function to boundary)**.** *Suppose F is a continuous function on $[-\pi, \pi]$ and $u = P[F]$. Then u extends continuously to $\overline{\mathbb{D}}$ and,*

$$\lim_{z \to e^{i\theta}} u(z) = F(\theta)$$

Notice that $P_r(\theta) > 0$, is even, 2π-periodic, $\int_{-\pi}^{\pi} P_r(\theta)\, d\theta = 2\pi$ and $\forall \delta > 0 \exists r_0,\ r > r_0 \implies \int_{|t|>\delta} P_r(\theta)\, d\theta < \delta$.

Proof. We show $\lim_{r \to 1} u(re^{i\theta}) = F(\theta)$.

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(\theta - t)P_r(t)\, dt$$

Then,

$$u(re^{i\theta}) - F(\theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} F(\theta - t)P_r(t)\, dt$$

and in particular,

$$|u(re^{i\theta}) - F(\theta)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} |F(\theta - t) - F(\theta)| P_r(t)\, dt = \frac{1}{2\pi} \left(\int_{|t|<\delta} + \int_{|t|>\delta} \right)$$

We show that these two integrals are bounded by $\frac{\varepsilon}{2}$. Since F is uniformly continuous, there exists a $\delta > 0$ such that,

$$|F(\theta - t) - F(\theta)| < \frac{\varepsilon}{2}$$

for $|t| < \delta$. Denote the two integrals by I and II, then,

$$|I| \leq \frac{1}{2\pi} \int_{|t|<\delta} \frac{\varepsilon}{2} P_r(t)\, dt \leq \frac{\varepsilon}{2}$$

Since F is continuous, it is bounded, $M = \|F\|_{L^\infty} < \infty$.

$$|II| \leq \frac{1}{2\pi} \int_{|t|>\delta} 2MP_r(t)\, dt$$

but by Poisson kernel properties, we can choose a value such that r is close to 1. □

Theorem 2.8. *Suppose $F(\theta) \in \mathcal{L}^p$ with $1 < p < \infty$ and $u = P[F]$. Set $F_r(\theta) = u(re^{i\theta})$, then $F_r \to F$ in L^p as $r \to 1$.*

Proof.

$$\|F_r(\theta) - F(\theta)\|_{L^p} = \left\| \frac{1}{2\pi} \int_{-\pi}^{\pi} (F(\theta - t) - F(\theta)) P_r(t) \right\|_{L^p} \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \|F(\theta - t) - F(\theta)\|_{L^p} P_r(t)\, dt \leq \frac{1}{2\pi} \left(\int_{|t|<\delta} + \int_{|t|>\delta} \right)$$

and we once again denote them I and II. $I < \frac{\varepsilon}{2}$ by the same claim as the last theorem. If $\|F\|_{L^p} = M$ then $\|F(\theta - t) - F(\theta)\|_{L^p} \leq 2M$ hence $II < \frac{\varepsilon}{2}$ by a choice of δ. □

3 Lecture 3 — 1.11.2026

3.1 Elementary study of the boundary behavior

We saw the theorem,

Theorem 3.1. *Suppose $F : \partial\mathbb{D} \to \mathbb{R}$ is in $\mathcal{L}^p(\partial\mathbb{D})$ and $u(re^{i\theta}) = P[F](re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(e^{it})P_r(\theta - t)\,dt$, is its Poisson extension to $\mathbb{D}$. If $F_r(e^{i\theta}) = u(re^{i\theta})$ then $F_r \to F$ in $\mathcal{L}^p(\partial\mathbb{D})$ as $r \to 1$.*

Remark. As stated, the theorem fails for $p = \infty$. If $v(z) = \operatorname{Re}\log z$ then $u(z) = v \circ m = \operatorname{Re}(\log(z) \circ m)$ for m the Möbius transformation mapping the unit disk onto the upper half plane $m : \mathbb{D} \xrightarrow{\sim} \mathbb{H}$. If $F_r = u(re^{i\theta})$ and if $F_r \to F$ in $\mathcal{L}^\infty$ then F would be continuous, but it is not as $u(-1)$ is non-continuity.

Theorem 3.2. *Suppose $F : [-\pi, \pi] \to \mathbb{R}$ is in $\mathcal{L}^\infty(\partial\mathbb{D})$ and $u(re^{i\theta}) = P[F](re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(e^{it})P_r(\theta - t)\,dt$, is its Poisson extension to $\mathbb{D}$. If $F_r(\theta) = u(re^{i\theta})$ then $F_r \to F$ in $\mathcal{L}^p(\partial\mathbb{D})$ in weak-$*$.*

Since $\mathcal{L}^\infty \simeq (\mathcal{L}^1)^*$, we need to check that,

$$\int_{-\pi}^{\pi} F_r(\theta)G(\theta)\,d\theta \xrightarrow{r\to 1} \int_{-\pi}^{\pi} F(\theta)G(\theta)\,d\theta$$

for all $G \in L^1(-[\pi, \pi])$.

Proof.

$$\mathcal{L}HS = \frac{1}{2\pi}\int_{-\pi}^{\pi}\int_{-\pi}^{\pi} F(t)P_r(\theta - t)\,dtG(\theta)\,d\theta \overset{P_r \text{ is even}}{=} \frac{1}{2\pi}\int_{-\pi}^{\pi}\int_{-\pi}^{\pi} F(t)P_r(t - \theta)\,dtG(\theta)\,d\theta$$

$G \in \mathcal{L}^1$ is bounded then $G(\theta)P_r(t - \theta)$ converges,

$$\to \int_{-\pi}^{\pi} F(t)G(t)\,dt$$

□

Theorem 3.3. *Suppose μ is a signed measure on $[-\pi, \pi]$,*

$$u(re^{i\theta}) = P[\mu](re^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - t)\,d\mu$$

is its Poisson extension to $\mathbb{D}$. If $F_r(\theta) = u(re^{i\theta})$, then $F_r(\theta)\,d\theta \to \mu$ in weak-$*$.

Remark. One generally does not have a total variation convergence.

Since $\mathcal{M} \simeq \mathcal{C}([-\pi, \pi])^*$, this means that,

$$\int_{-\pi}^{\pi} F_r(\theta)G(\theta)\,d\theta \to \int_{-\pi}^{\pi} G(\theta)\,d\mu(\theta)$$

as $r \to 1^-$, for every continuous function G.

Proof.

$$\mathcal{L}HS = \frac{1}{2\pi}\int_{-\pi}^{\pi}\int_{-\pi}^{\pi} P_r(t - \theta)\,d\mu(t)\,G(\theta)\,d\theta \xrightarrow{r\to 1 \text{ in } \mathbb{C}} G(t)$$

As $r \to 1^-$, $\mathcal{L}HS \to \int_{-\pi}^{\pi} G(t)\,d\mu(t)$. □

Theorem 3.4 (Fatou's theorem)**.** *Suppose μ is a signed measure on $[-\pi, \pi]$ and $\eta(t) = \int_{-\pi}^{t} d\,\mu$ be its distribution function. If $\eta'(t)$ exists, then $P[\mu](z) \to \eta'(t)$ as $z \to e^{it}$ non-tangentially, namely the sequence of points is bounded by its angle from the point of convergence.*

Since monotone functions are differentiable almost everywhere, the claim holds for almost every $t \in [-\pi, \pi]$. If $\mu = g(t)\,dt$ with g continuous then $\eta'(t) = g(t)$.

Recall that $P_r(t) = \frac{1-r^2}{1+r^2-2r\cos t}$. Set,

$$F_r(t) = -\frac{1+r}{2r} \cdot tP_r'(t) = \frac{(1+r)^2(1-r)t\sin t}{(1+r^2-2r\cos t)^2}$$

Lemma 3.5. *$F_r > 0$, even, $\frac{1}{2\pi}\int_{-\pi}^{\pi} F_r(t)\,dt = 1$, and for every $\delta, \varepsilon > 0$ there is $0 < r_0 < 1$ such that for any $r_0 < r < 1$,*

$$\int_{[-\delta,\delta]^C} F_r(t)\,dt < \varepsilon$$

The reason for the multiplication by $\frac{1+r}{2r}$ is that,

$$\int_{-\pi}^{\pi} tP'(t)\,dt = tP(t)\,|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} P(t)\,dt = 2\pi \cdot \frac{1-r}{1+r} - 2\pi = 2\pi \cdot \frac{-2r}{1+r}$$

so that they will cancel out. Without loss of generality, we may assume that $\theta = 0$, Our claim is that $\lim_{r\to 1} u(re^{i\theta}) = \eta'(0)$ where $u = P_\mu$.

$$\begin{aligned} P_\mu(r) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1-r^2}{1+r^2-2r\cos t}\,d\mu(t) \\ &= \frac{1}{2\pi}\,\frac{1-r^2}{1+r^2-2r\cos t}\eta(t)\Big|_{-\pi}^{\pi} - \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\partial P_r}{\partial t}\cdot\eta(t)\,dt \\ &= \frac{1}{2\pi}\frac{1-r}{1+r}\cdot\mu([-\pi,\pi]) + \frac{1}{2\pi}\frac{2r}{1+r}\int_{-\pi}^{\pi} -\frac{1+r}{2r}\cdot t\cdot\frac{\partial P_r}{\partial t}\cdot\frac{\eta(t)}{t}\,dt \end{aligned}$$

Denote the terms in the last formula I and II. As $r \to 1^-$, $I \to 0$.

$$\frac{2r}{1+r^2} \to 1$$

and,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} F_r(t)\cdot\frac{\eta(t)}{t}\,dt$$

$\frac{\eta(t)}{t}$ is continuous at 0 by our assumption, we also assume without loss of generality that $\eta(0) = 0$ by subtracting a constant value, then,

$$\lim_{t\to 0, t\neq 0}\frac{\eta(t)}{t} = \eta'(0)$$

Summary: $\eta(t) = \int_{-\pi}^{t} d\,\mu = \mu([-\pi,t])$. Pick $t \in [-\pi,\pi]$ where η' exists. $u(z)$ is harmonic then $u(e^{i\alpha}z)$ is also harmonic, therefore we can rotate it such that we choose $t = 0$. We want $\eta(0) = 0$, replace μ by $\mu - c\,dm$ where $c \in \mathbb{R}$ is chosen that $\mu([-\pi,0]) = c\cdot m([-\pi,0]) = c\pi$. $P_\mu(r) = I + II$. If $\eta(0) = 0$ and $\eta'(0)$ exists then $g(t) = \frac{\eta(t)}{t}$ is continuous at 0, therefore $II \to \eta'(0)$.

Theorem 3.6 (F & M. Riesz)**.** *Suppose μ is a signed measure on $[-\pi,\pi]$. If,*

$$\forall n \in \mathbb{N}, \quad \hat{\mu}(-n) := \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{in\theta}\,d\mu = 0$$

then μ is absolutely continuous with respect to m.

Definition 3.7 (Function created measure)**.** If μ is a measure and $f \in \mathcal{L}^1(\mu)$, then $(f\,d\mu)(E) = \int_E f\,d\mu$.

Proof. Step 0: Suppose not. Then $\mu = \mu_{ac} + \mu_s$ where $\mu_s \neq 0$. There is E with $m(E) = 0$ but $\mu(E) = \mu_s(E) \neq 0$. If μ is not absolutely continuous, then neither is $e^{i\theta}\,d\mu$, then there is E such that $m(E) = 0$ but $\int_E e^{i\theta}\,d\mu \neq 0$. By inner regularity, we may assume that E is closed. Denote the complementary arcs by $\{I_j\}$. So, $\partial\mathbb{D} \setminus E = \bigcup I_j$, as E is closed.

Step 1: There exists a function $F : \partial\mathbb{D} \to (0,\infty]$ which satisfies,

- $F \in \mathcal{L}^1(\partial\mathbb{D})$
- $F \in C^\infty(\partial D \setminus E)$
- $F(e^{i\theta}) \to \infty$ as $e^{i\theta} \to E$

$l_j = m(I_j)$ then $\sum l_j = 2\pi$. There exists $m_j \nearrow \infty$ such that $\sum l_j m_j < \infty$. On I_j we construct F such that $F(t) \geq m_j$ then $\int_{I_j} F(t)\, dt \geq l_j \cdot m_j$. Let $u = P[F]$ and $\tilde{u}$ its harmonic conjugate, $u + i\tilde{u}$ is analytic function with positive real part. A fact we will accept is that $\tilde{u} \in C(\partial\mathbb{D} \setminus E)$.

Step 2: The disk algebra $\mathcal{A}$ consists of all analytic functions $\mathbb{D}$ which extend continuously to $\mathbb{D}$, $\mathcal{A} = \mathrm{Hol}(\mathbb{D}) \cap C(\bar{\mathbb{D}})$. Then $\phi = \frac{u+i\tilde{u}}{1+u+i\tilde{u}}$ is a disk algebra function such that $|\phi(z)| < 1$ on $\mathbb{D} \setminus E$, $\phi(z) = 1$ "peak set".

Step 3: We claim that,

$$\int_{-\pi}^{\pi} \phi(e^{i\theta})^k \cdot e^{i\theta}\, d\mu = 0$$

for any $k \in \mathbb{N}$. We use a trick to observe ϕ in the unit disk instead of the boundary, in which is analytic,

$$\int_{-\pi}^{\pi} \phi(e^{i\theta})^k \cdot e^{i\theta}\, d\mu = \lim_{r\to 1} \int_{-\pi}^{\pi} \phi(re^{i\theta})^k \cdot re^{i\theta}\, d\mu$$

so $\phi(z)^k = a_0 + a_1 z + \cdots$ converges absolutely and uniformly on compact subsets of $\mathbb{D}$. It is possible to integrate term by term and recall that $\int e^{in\theta}\, d\mu = 0$ for $n \geq 0$.

Step 4: By dominated convergence theorem,

$$0 = \int_{-\pi}^{\pi} \phi(e^{i\theta})^k e^{i\theta}\, d\mu \xrightarrow{k\to\infty} \int_E e^{i\theta}\, d\mu(\theta)$$

This is a contradiction to our assumption that $\int_E e^{i\theta}\, d\mu(\theta) \neq 0$. □

Recall that H^p consists of all analytic functions on $\mathbb{D}$ such that,

$$\sup_{0<r<1} \int_{-\pi}^{\pi} |f(er^{i\theta})|^p\, d\theta < \infty$$

If $p > 1$, there exists a function $F : \partial\mathbb{D} \to \mathbb{R}$ in $\mathcal{L}^p$ such that $f(z) = P[F]$. If $p = 1$, then there is a measure $\mu \in \mathcal{M}(\partial\mathbb{D})$ such that $f = P[\mu]$. Brothers Riesz: $\mu = F\, dm$ for some $F \in \mathcal{L}^1$, if,

$$\int_{\partial\mathbb{D}} z^n\, d\mu = 0$$

for $n \in \mathbb{N}$. $f_r \to \mu$ in weak-$*$.

$$\int_{\partial\mathbb{D}} z^n f(rz)\, dm \to \int_{\partial\mathbb{D}} z^n\, d\mu$$

4 Lecture 4 — 8.11.2026

Definition 4.1 (Hardy space)**.** The harmonic Hardy space h^p is the set of harmonic functions u on $\mathbb{D}$ satisfying,

$$\|u\|_{h^p}^p = \sup_{0<r<1} \int_{-\pi}^{\pi} |u(re^{i\theta})|^p \, d\theta < \infty$$

For $p > 1$ we saw that $h^p \simeq \mathcal{L}^p(\partial\mathbb{D})$,

$$P : \mathcal{L}^p \xrightarrow{\sim} h^p, \qquad F(\theta) \mapsto \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - t) F(t) \, dt$$

For $p = 1$ this is false, if $u(re^{i\theta}) = \frac{1-r^2}{1+r^2-2r\cos\theta}$. In this case $\mathcal{L}^1 \subsetneq h^1 \subseteq \mathcal{M}$.

Lemma 4.2. *Suppose $F \in C^1([-\pi, \pi])$ and suppose,*

$$u(er^{i\theta}) = P[F](er^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1 - r^2}{1 + r^2 - 2r\cos(\theta - t)} F(t) \, dt$$

We have seen that,

$$\tilde{u}(re^{i\theta}) = Q[F](re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{2r\sin s}{1 + r^2 - 2r\cos s} F(\theta - s) \, ds$$

Then $\tilde{u}$ extends to a continuous function on $\bar{\mathbb{D}}$.

We know that $u \to \tilde{u}$ preserves $\mathcal{L}^p$ for $1 < p < \infty$, C^α for $0 < \alpha < 1$, $W^{s,p}$ for $p > 1$, Z, BMO etc. And we define,

$$\|f\|_{C^\alpha} = \|f\|_\infty + \sup_{x,y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}$$

This defines Hölder space.

Proof of the lemma.

$$\tilde{u}(re^{i\theta}) = \frac{1}{\pi} \int_0^{\pi} \frac{r\sin s}{1 + r^2 - 2r\cos s} (F(\theta - s) - F(\theta + s)) \, ds$$

but $r\sin s = 2r\sin\frac{s}{2}\cos\frac{s}{2}$ and $1+r^2-2r\cos s = (1-r)^2+2r-2r\cos s = (1-r)^2+2r(1-\cos s) = (1-r)^2+rr\sin^2\frac{s}{2}$, then,

$$= \frac{1}{\pi} \int_0^{\pi} \frac{2r\sin\frac{s}{2}\cos\frac{s}{2}}{(1-r)^2 + 4r\sin^2\frac{s}{2}} (F(\theta - s) - F(\theta + s)) \, ds$$

As $r \to 1^-$, $\frac{2r\sin\frac{s}{2}\cos\frac{s}{2}}{(1-r)^2+4r\sin^2\frac{s}{2}}$ tends to $\frac{1}{2\tan\frac{s}{2}}$. Therefore,

$$\lim_{r\to 1} \tilde{u}(re^{i\theta}) = \frac{1}{\pi} \int_0^{\pi} \frac{F(\theta - s) - F(\theta + s)}{2\tan\frac{s}{2}} \, ds$$

Since $F \in C^1$, this integral converges absolutely. I.E.,

$$\int \left| \frac{F(\theta - s) - F(\theta + s)}{2\tan\frac{s}{2}} \right| ds < \infty$$

To justify the passage to the limit, we use Lebesgue's dominated convergence theorem,

$$\frac{C}{\pi} \int \frac{|F(\theta - s) - F(\theta + s)|}{\tan\frac{s}{2}} < \infty$$

□

Theorem 4.3 (Uniqueness for H 1 functions)**.** *Suppose $F \in H^1$. If F has non-tangential boundary value 0 on a set of positive Lebesgue measure E, then $F \equiv 0$.*

Proof. Without loss of generality, we may assume that $F(0) \neq 0$ ($F(z) = z^k G(z)$ such that $G(0) \neq 0$). Let $u = P[\frac{1}{|E|}\chi_E - \frac{1}{|E^C|}\chi_{E^C}]$, then $u(0) = \lim_{r\to 1} \frac{1}{2\pi}\int_{-\pi}^{\pi} u(re^{i\theta}) \, d\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} u(e^{i\theta}) \, d\theta$. Denote $\tilde{u}$, the harmonic conjugate of u, set $\varphi(z) = e^{u(z)+i\tilde{u}(z)}$, a holomorphic function and $\varphi(0) = 1$. $\sup_{z\in\mathbb{D}} |\varphi(z)| < \infty$ therefore $\varphi \in H^\infty$. Since φ is bounded, $\varphi^k F \in \mathbb{H}^1$ for every integer k.

$$F(0) = \varphi^k(0)F(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \varphi^k(e^{i\theta}) F(e^{i\theta}) \, d\theta = \frac{1}{2\pi} \int_{E^C} \varphi^k(e^{i\theta}) F(e^{i\theta}) \, d\theta \xrightarrow{k\to\infty} 0$$

□

Theorem 4.4 (Privalov). *Suppose F is an analytic function in $\mathbb{D}$. If F has non-tangential boundary value 0 on a set of positive Lebesgue measure, then $F \equiv 0$.*

Remark. There exists $F \not\equiv 0$ analytic on $\mathbb{D}$ such that the radial boundary value is 0 for almost every θ.

We take for a fact the Riemann mapping theorem, that there exists a conformal mapping $\varphi : \mathbb{D} \to \Omega$ for $\Omega \subsetneq \mathbb{C}$ such that Ω is s.c.

Theorem 4.5 (Carathéodory). *If Ω is a Jordan domain, then φ extends to a homeomorphism from $\bar{\mathbb{D}}$ to $\bar{\Omega}$.*

Definition 4.6 (Rectifiable curves). if $\gamma : I \to \mathbb{C}$ is a curve, then we define length $\gamma = \sup_P \sum_{i=0}^{n-1} |z_{i+1} - z_i|$. We say that γ is rectifiable if length $\gamma < \infty$.

Another fact we will accept,

Proposition 4.7. *If γ is rectifiable, it has an arclength parameterization, $\gamma : [0, L] \to \mathbb{C}$ such that,*

$$\text{length}\, \gamma([0,t]) = t$$

Definition 4.8 (Lebesgue measure of a curve). γ is defined as $\gamma_* m \mid_{[0,L]}$. If $\gamma(t) = (x(t), y(t)) \in C^1$ then,

$$\text{length}\, \gamma = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}$$

Definition 4.9 (Subharmonic & Superharmonic function). We say that $u : \Omega \to \mathbb{R}$ is subharmonic if it is upper semi-continuous and satisfies the sub mean-value property,

$$u(z_0) \le \frac{1}{2\pi} \int_{-\pi}^{\pi} u(z_0 + re^{i\theta})\, d\theta$$

A function is called superharmonic if it satisfies the super mean-value property, which is the exact same property with the relation inverted.

Then $\limsup_{z \to z_0} u(z) \le u(z_0)$. Subharmonic functions form a cone, the multiplication of a subharmonic function u and $\lambda > 0$ is subharmonic, and the addition of subharmonic functions is harmonic.

Remark. If u is subharmonic then $-u$ is superharmonic. If f is analytic then $|f|$ is subharmonic,

$$|f(z_0)| = \left| \frac{1}{2\pi} \int f(z_0 + re^{i\theta})\, d\theta \right| \le \frac{1}{2\pi} \int |f(z_0 + re^{i\theta})|\, d\theta$$

Subharmonic functions satisfy the maximum principle,

$$\forall z_0 \in \Omega,\ u(z_0) \le \limsup_{z \to \partial\Omega} u(z)$$

If equality holds, then u is constant.

Remark. Superharmonic functions satisfy the minimum principle, harmonic functions satisfy both principles.

Theorem 4.10 (F & M Riesz). *Suppose Ω is a Jordan domain. $\partial\Omega$ is rectifiable iff $\varphi' \in H^1$.* length$(\partial\Omega) = \|\varphi'\|_{H^1}$.

Proof. $\Longleftarrow$: By Carathéodory's theorem, φ extends to a homeomorphism of the closures.

$$\begin{aligned}\sum_{k=0}^{n-1} |\varphi(e^{i\theta_{k+1}}) - \varphi(e^{i\theta_k})| &= \lim_{r\to 1} \sum_{k=0}^{n-1} |\varphi(re^{i\theta_{k+1}}) - \varphi(re^{i\theta_k})| \\ &= \lim_{r\to 1} \sum_{k=0}^{n-1} \left| \int_{\theta_k}^{\theta_{k+1}} \varphi'(re^{i\theta}) ire^{i\theta}\, d\theta \right| \le \lim_{r\to 1} \sum_{k=0}^{n-1} \int_{-\pi}^{\theta_\pi} |\varphi'(re^{i\theta})|\, d\theta = \|\varphi'\|_{H^1}\end{aligned}$$

$\Longrightarrow$: Fix a partition of $[0, 2\pi]$, $0 = \theta_0 < \cdots \theta_n = 2\pi$. Define,

$$S(z) = \sum_{k=0}^{n-1} |\varphi'(e^{i\theta_{k+1}} z) - \varphi'(e^{i\theta_k} z)|$$

This is a subharmonic function, by the maximum principle,

$$S(z_0) \leq \max_{|\zeta|=1} S(\zeta) \leq \text{length}\, \partial\Omega$$

so length$(\varphi(|z| = r)) \leq$ length$(\partial\Omega)$, but this length is exactly $\int_{-\pi}^{\pi} |\varphi(re^{i\theta})ire^{i\theta}|\, d\theta$. □

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